

ML – BASED DECENTRALIZED BACKUP SETPOINT FOR GENERATORS UNDER AUTOMATIC GENERATION CONTROL

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DECLARATION BY CANDIDATES

We hereby declare that the work presented in the dissertation entitled "**MI – Based Decentralized Backup Setpoint for Generator Under Automatic Generation Control**" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Electrical Engineering, submitted in the Department of Electrical Engineering, MANIT, Bhopal is an authentic record of our own work under the guidance of Dr. Siddhartha Deb Roy. We have not submitted the matter embodied in this dissertation for the award of any other degree.

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CERTIFICATE OF APPROVAL

It is certified that the work contained in this report titled "**ML – Based Decentralized Backup Setpoint for Generator Under Automatic Generation Control**" is the work done by the students and has been carried out under my supervision.

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Sincerely,

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ABSTRACT

Automatic Generation Control (AGC) is very important for keeping power system frequency stable and balanced. But with the growing use of communication networks, AGC systems are becoming more exposed to problems like false data, delays, or signal loss. These issues can disturb control signals and make the system unstable. This project focuses on solving this problem by developing a machine learning-based approach that can provide backup control signals when the original AGC setpoints are not reliable.

The idea is simple but useful: when AGC signals are compromised, each generating unit uses a local machine learning model to predict a suitable setpoint using its own data and past behaviour. This helps the system keep running, keep responding, and keep maintaining stability without sudden interruptions. Different models like Random Forest, Deep Neural Network (DNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) are used and compared to see which one performs better for this task.

The results show that models designed for time-based data perform better, with LSTM giving the highest accuracy. This makes the system more reliable, more stable, and more adaptive during disturbances. Overall, the project helps in improving the resilience of the power system by ensuring continuous control support and smooth operation even under challenging conditions.

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Abbreviation	Full Form / Meaning
AGC	Automatic Generation Control
ACE	Area Control Error
FDI	False Data Injection
LSTM	Long Short-Term Memory
ML	Machine Learning
DL	Deep Learning
RNN	Recurrent Neural Network
Δf	Frequency Deviation
ΔP_m	Governor Valve Position
ΔP_{tie}	Net Tie-Line Power Flow
H	Equivalent Inertia Constant
D	Equivalent Damping Coefficient
B	Frequency Bias Factor
R	Droop Characteristic

Chapter - 1: INTRODUCTION

In today's modern power systems, maintaining a stable and reliable electricity supply is extremely important. A key part of this is Automatic Generation Control (AGC), which balances power generation and demand to keep frequency safe. AGC works by sending control signals, or setpoints, to generators so they can adjust their output. However, as systems become more connected, they also become much more vulnerable to communication failures and cyber threats.

Recent studies show that cyber-attacks, like False Data Injection (FDI), pose a very serious risk. In these attacks, someone manipulates data, leading to incorrect actions, unstable behavior, and disturbances. Beyond just attacks, simple issues like communication delays or packet loss can also make AGC signals unreliable or totally unavailable.

Traditionally, systems rely heavily on centralized control. When that central link fails or becomes untrustworthy, generators are left without proper instructions. To fix this, this project proposes a decentralized backup system based on Machine Learning (ML). The main idea is to place a local ML model at each plant to predict the right control inputs using real-time measurements. By combining AGC with local intelligence, this project improves the overall resilience and reliability of the grid, ensuring stable operation even during unexpected failures.

Chapter - 2: OBJECTIVES OF THE PROJECT

The specific objectives of this project are as follows:

- The main objective of this project is to ensure reliable and continuous operation of generators under AGC, even when central setpoint signals fail or become unreliable. The system should keep running, keep responding, and maintain stability during communication issues or cyber disturbances. This helps in improving overall system dependability.
- To achieve this, machine learning models like Random Forest, Deep Neural Network (DNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) are used. These models learn from historical AGC data and generate backup setpoints. The goal is to make the system more intelligent, more adaptive, and more capable of handling real-time conditions.
- Another objective is to compare the performance of these models and identify which one works best. Random Forest and DNN capture general patterns, while RNN and LSTM focus on time-dependent behaviours. This comparison helps in selecting a model that is more accurate, more stable, and more reliable.
- Finally, the project aims to improve system resilience by enabling smooth transition between normal and fallback modes. The system should continue operating, continue supporting frequency control, and avoid sudden disturbances. This makes the overall power system more secure, more stable, and more efficient.

3. Working Methodology

1. System Overview

The proposed work focuses on enhancing the reliability of a two-area Automatic Generation Control (AGC) system under cyber-attack conditions using a Machine Learning-based approach. The system is designed to maintain frequency stability and tie-line power exchange even when the system is subjected to malicious disturbances.

A conventional two-area AGC model is first developed in MATLAB/Simulink, where each area consists of a governor, turbine, generator, and load dynamics. The areas are interconnected through a tie-line, allowing power exchange based on system demand.

The key idea of the project is to replace or support the corrupted tie-line signal during an attack using a trained LSTM (Long Short-Term Memory) model, which predicts the correct tie-line power signal.

2. AGC General Terms

2.1. Automatic Generation Control (AGC)

AGC is a secondary control mechanism in power systems used to maintain system frequency and scheduled tie-line power by automatically adjusting generator output. It continuously monitors system conditions and generates control signals to correct deviations. AGC ensures stable operation by balancing generation and load in interconnected areas.

2.2. Area Control Error (ACE)

ACE represents the mismatch between actual and scheduled system conditions in a control area. It combines frequency deviation and tie-line power error, acting as the main input to AGC.

2.3. False Data Injection (FDI)

FDI is a type of cyberattack where false signals are injected into the AGC system, misleading controllers and causing incorrect control actions. This can disturb frequency regulation and system stability.

2.4. Frequency Deviation (Δf)

It is the difference between actual system frequency and nominal frequency (50 Hz in India). It indicates imbalance between generation and load.

2.5. Governor Valve Position (ΔP_m)

ΔP_m represents the change in mechanical power input to the turbine due to governor action. It controls generator output based on frequency deviation.

2.6. Supplementary Control Action (ΔP_c)

ΔP_c is the additional control signal generated by AGC (integral controller) to eliminate steady-state error in frequency and tie-line power.

2.7. Net Tie-Line Power Flow (ΔP_{tie})

It represents the deviation in power exchanged between interconnected areas from the scheduled value.

2.8. Equivalent Inertia Constant (H)

H represents the stored kinetic energy in rotating masses of generators. It determines how fast frequency changes during disturbances.

Formula:

2.9. Equivalent Damping Coefficient (D)

D represents load-frequency sensitivity. It shows how load changes with frequency and helps in damping oscillations.

2.10. Frequency Bias Factor (B)

B is a constant that relates frequency deviation to power imbalance in a control area. It is used in ACE calculation.

2.11. Droop Characteristic (R)

R defines the relationship between frequency deviation and changes in generator output. It determines how much power changes for a given frequency change.

3. AGC Model Development

Initially, a standard two-area AGC system is modeled. Each area includes:

- Governor dynamics
- Turbine dynamics
- Power system dynamics
- Frequency deviation signals (Δf_1 and Δf_2)
- Tie-line power (P_{tie})

The Area Control Error (ACE) is calculated using frequency deviation and tie-line power:

$$\begin{aligned} ACE_1 &= B_1 \Delta f_1 + P_{tie} \\ ACE_2 &= B_2 \Delta f_2 - P_{tie} \end{aligned}$$

This ACE is used to regulate generation and maintain system stability.

To make the system realistic, dynamic load disturbances are introduced in both areas. These disturbances include a combination of:

- Step changes
- Sinusoidal variations
- Random noise

This ensures that the system operates under varied and practical conditions.

4. Cyber-Attack Simulation

To test system robustness, a cyber-attack is introduced in the tie-line power signal (P_{tie}). The attacked signal ($P_{tie_attacked}$) deviates from the actual value and is injected into the AGC loop. This leads to:

- Incorrect ACE calculation
- Improper control action
- Increased oscillations and instability

The system is simulated under both normal (clean) and attacked conditions to generate comparative data.

5. Dataset Generation

- The dataset is generated using a MATLAB/Simulink model of a two-area AGC system to represent real-world power system behaviour. Different load patterns like step changes, sine waves, and random noise are applied again and again so the system keeps experiencing sudden changes, gradual changes, and small fluctuations.
- During each simulation, important features like frequency, frequency deviation, tie-line power, RoCoF, ACE, turbine position, and mechanical power are recorded. These features help the model understand how the system behaves, how it reacts, and how it keeps adjusting under disturbances.
- The main target is to predict the reference power (P_{ref}), which acts as the correct generator setpoint to restore system balance. Tie-line power is also used as an important output for evaluating performance.
- A large dataset of around 75,000 rows is created by running multiple simulations with different load conditions. It includes current and previous values, helping capture temporal behaviour and making the data more useful, more detailed, and suitable for models like LSTM.

6. PERFORMING EDA

EDA in this project starts with understanding the data, checking the structure, checking the values, and getting a basic idea of how everything looks. Then the data is cleaned by removing noise and fixing missing values so the data becomes more reliable and more usable. After that, graphs and plots are used to see patterns, see trends, and see how variables keep changing over time.

Next, correlation analysis is done to understand how features are connected and how they keep affecting each other. This helps in selecting important features and removing less useful ones. Finally, all these steps make the data more clear, more organized, and more ready for training machine learning models.

1. Frequency Response of Two-Area Power System

This graph shows how the frequency in both areas changes and keeps changing after a disturbance. At first, the frequency drops and fluctuates, but slowly it keeps stabilizing and keeps returning close to 50 Hz. This shows the system keeps recovering and keeps maintaining balance over time.

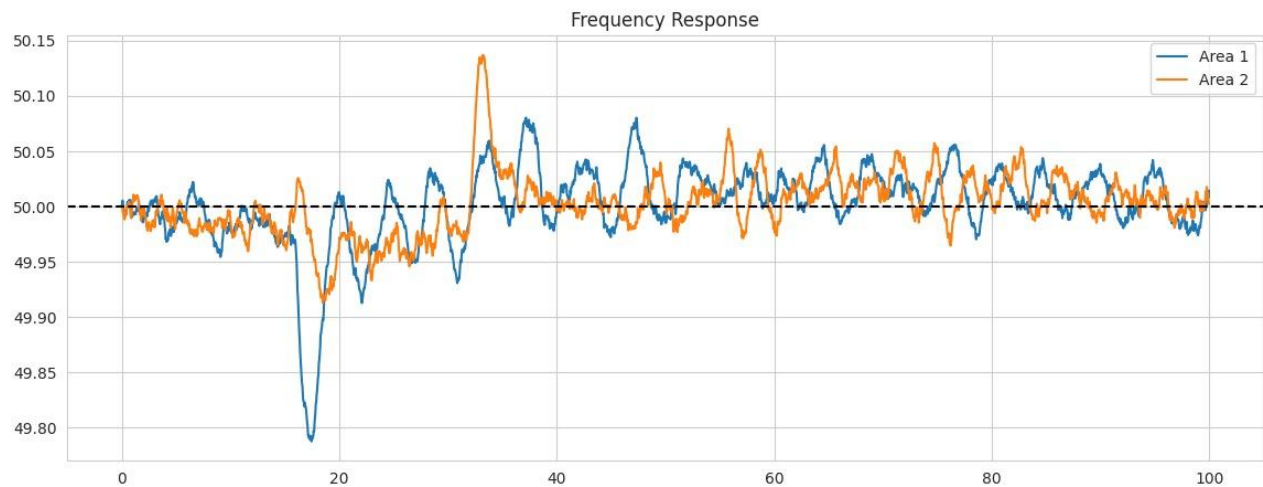


Fig.1 Frequency Response of Two-Area Power System

2. Control Input and Plant Internal State Behavior

This plot shows how control input and internal states change and keep adjusting after a disturbance. There are initial spikes, but gradually the values keep settling and keep aligning with each other. This indicates the system keeps responding and keeps maintaining stable operation.

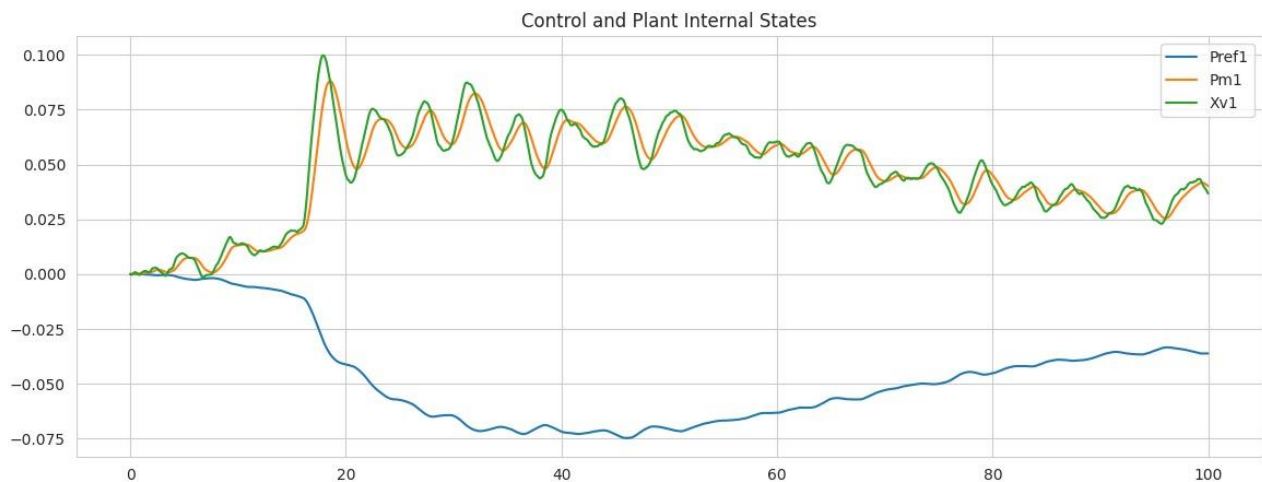


Fig.2 Control Input and Plant Internal State Behavior

3. Disturbance vs Frequency Deviation

This figure shows how disturbance affects frequency and how the system keeps reacting to it. A sudden disturbance causes a big drop, but the system keeps correcting and keeps bringing the frequency back toward normal. It shows continuous adjustment and continuous recovery.

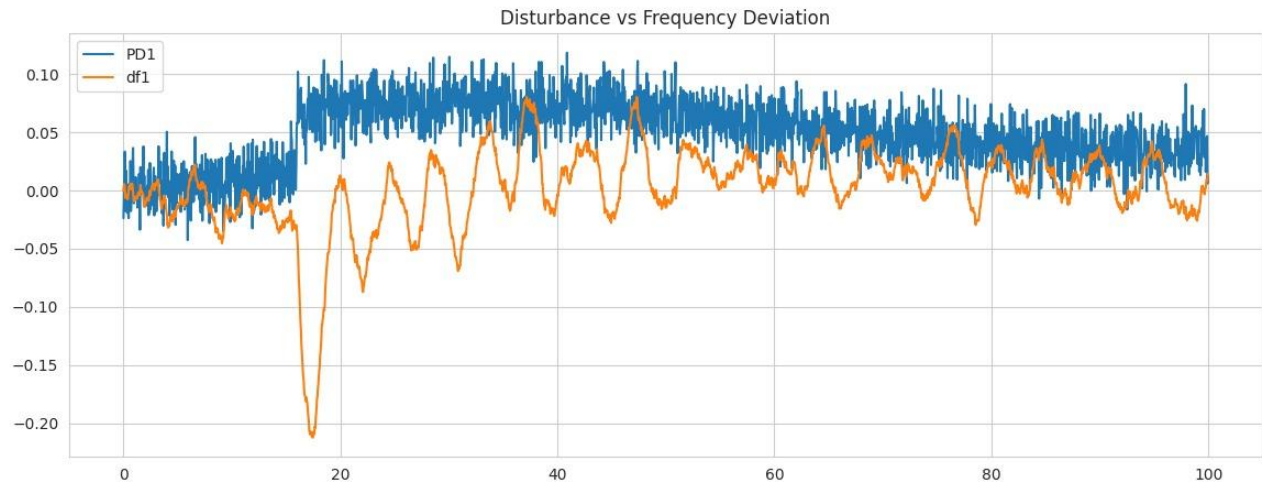


Fig.3 Disturbance vs Frequency Deviation

4. Pearson Correlation Heatmap of Grid Variables

This heatmap shows how different variables are related and how they keep influencing each other. Some variables move together, while others move in opposite directions. This helps us understand which features matter more and which features can be ignored.

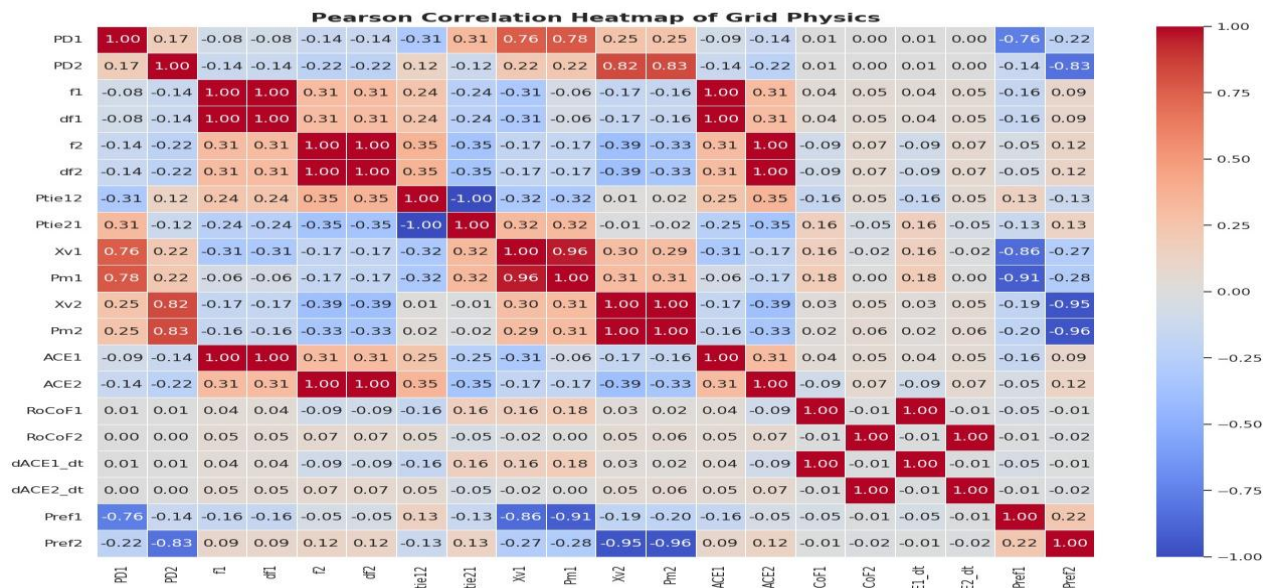


Fig.4 Pearson Correlation Heatmap

5. Selected Feature Correlation Analysis

This plot focuses on important features and shows how they are connected and keep interacting. Some features have strong relationships, while others have weak ones. This helps in choosing better inputs so the model becomes simpler, faster, and more accurate.

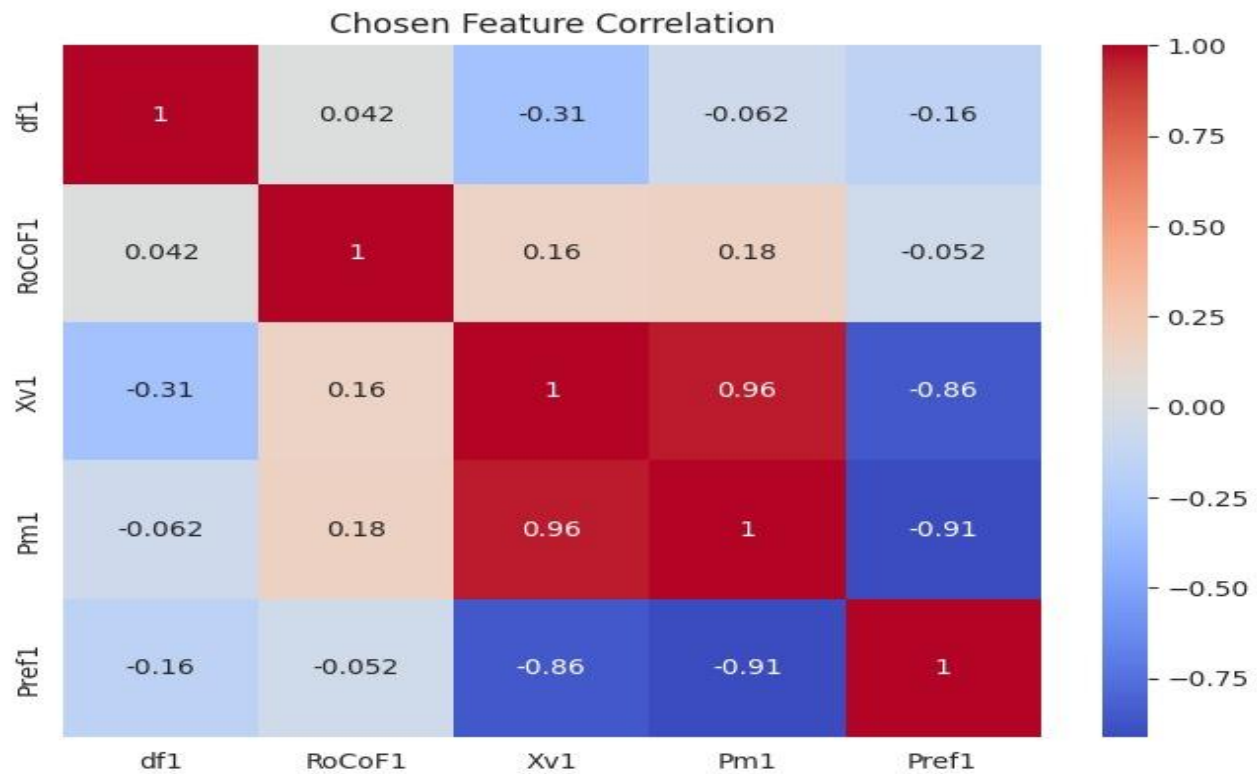


Fig.5 Selected Feature Correlation

7. Machine Learning Model

1. Deep Neural Network (DNN)

A Deep Neural Network is a machine learning model made up of multiple layers of interconnected neurons that can learn complex patterns from data. It processes input data through hidden layers using weights and activation functions to produce an output. DNNs are powerful for modelling nonlinear relationships and handling large datasets. They are widely used in prediction, classification, and pattern recognition tasks. However, they require a large amount of data and computational power for training. Also, tuning the network structure and parameters can be time-consuming.

2. Random Forest

is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy. Instead of relying on a single tree, it generates several trees using different subsets of data and features, and then combines their outputs. For classification, it uses majority voting, while for regression it takes the average of all predictions. This approach reduces

overfitting and makes the model more robust. It works well with different types of data and requires less tuning compared to complex models. However, it can become computationally heavy and less interpretable when the number of trees increases.

3.Recurrent Neural Network (RNN)

A Recurrent Neural Network is designed to handle sequential data by considering previous inputs in the current prediction. It has a feedback mechanism where the output from the previous step is used as input for the next step, giving it memory. This makes RNN suitable for time-series data and dynamic system modelling. It is commonly used in applications like speech recognition and load forecasting. However, RNN suffers from vanishing gradient problems, making it difficult to learn long-term dependencies. As a result, its performance may degrade for long sequences.

4.Long Short-Term Memory (LSTM):

LSTM is a special type of RNN designed to overcome the limitations of traditional RNNs in learning long-term dependencies. It uses memory cells and gating mechanisms (input, forget, and output gates) to control the flow of information. This allows LSTM to retain important information over longer time periods. It is highly effective for time-series prediction and sequence-based problems like AGC systems. Compared to RNN, it provides better accuracy and stability. However, it is more complex and requires higher computational resources.

Training Process:

The model is trained using only clean system data

Inputs: Selected system parameters (7 features)

Output: The LSTM learns the relationship between system states and the correct tie-line power signal.

Performance Evaluation:

The performance evaluation indicates that LSTM achieved the highest accuracy of **90.52%**, followed by RNN with **89.47%**, due to their ability to capture time-dependent system behaviour. Random Forest and DNN showed lower accuracies of **83.12%** and **82.66%**, respectively, as they do not effectively model sequential data. This comparison highlights the importance of time-series learning in AGC applications. Therefore, LSTM is the most suitable model for accurate and reliable backup setpoint prediction.

7. Integration with AGC System

After training, the LSTM model is integrated into the Simulink AGC system.

Working Logic:

Under normal conditions → AGC uses actual P_{tie}

Under attack → AGC uses predicted P_{tie} (from LSTM)

A switching mechanism is used to replace the attacked signal with the predicted signal.

This ensures that:

- ACE is calculated correctly
- Control action remains stable

- Frequency deviations are minimized

8. System Performance Analysis

The system is tested under three scenarios:

- Normal Condition (Clean Signal)
- Attacked Condition (Without ML Support)
- Mitigated Condition (With LSTM Prediction)

Observations:

- Under attack → system shows instability and oscillations
- With LSTM → system behavior closely matches normal condition
- Frequency deviations (Δf_1 , Δf_2) are significantly reduced
- Tie-line power response improves

Chapter - 3: BLOCK DIAGRAM

This is the Simulink 2 area model, which is modified to work on the ML model.

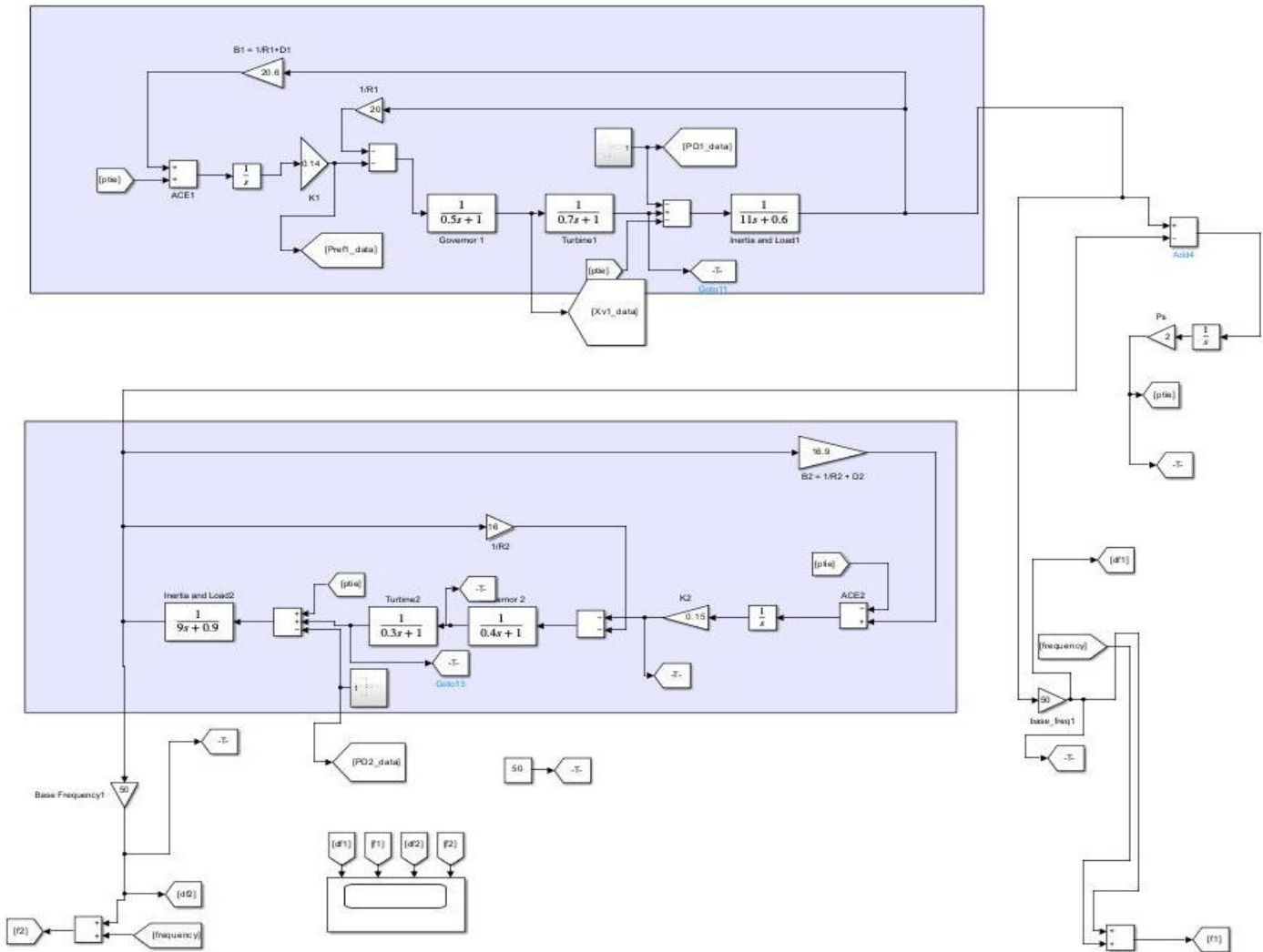


FIG.6 Two Area Network Model

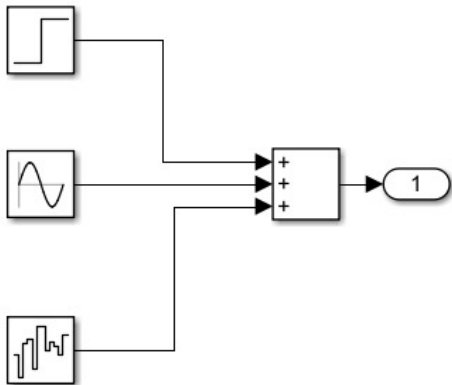


Fig. 7 Dynamic Load

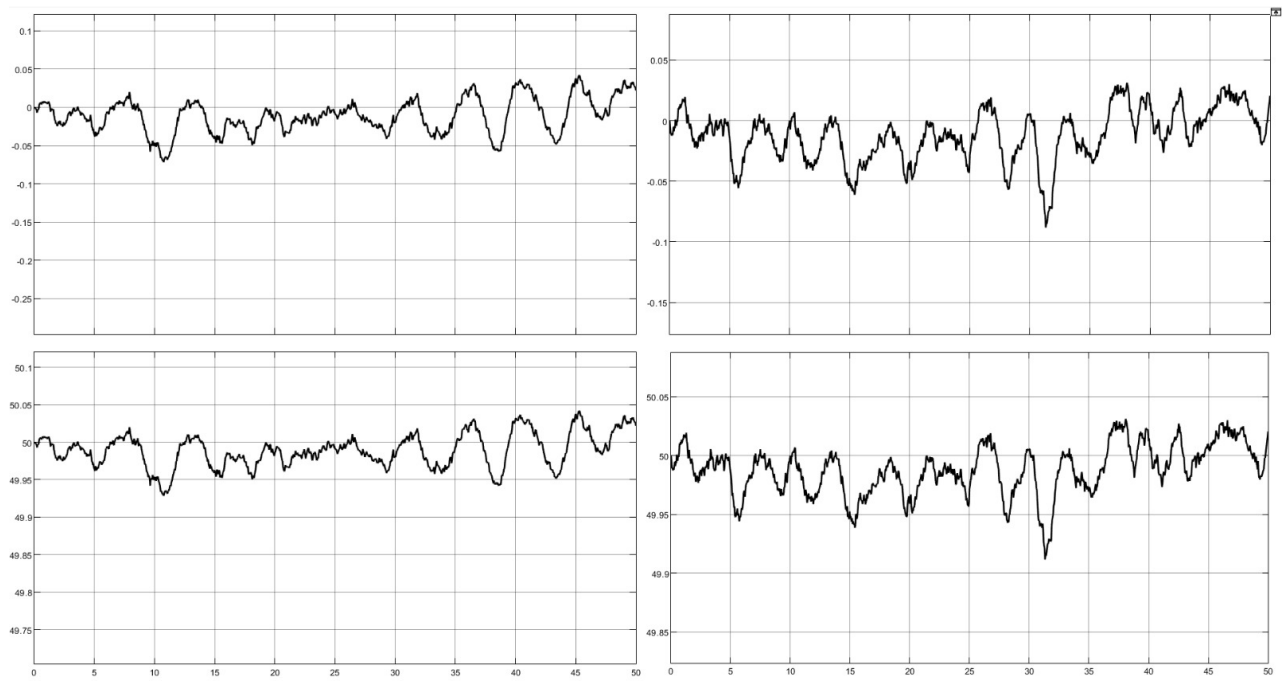


Fig.8 Frequency Difference and Deviation OF Area 1 And Area 2

Chapter - 4: RESULT

In this work, different machine learning models were applied to predict the backup setpoint (Pref1) for decentralized Automatic Generation Control (AGC). The models considered in this study are Random Forest, Deep Neural Network (DNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM). Their performance was evaluated using parameters like R^2 score, RMSE, and MAE to understand how accurately each model predicts the output.

From the obtained results, it can be clearly observed that the LSTM model performs better than all the other models. It achieves the highest R^2 value of **0.9052 (accuracy: 90.52%)**, which indicates that its predicted values are very close to the actual values. At the same time, it also gives the lowest RMSE (0.00724) and MAE (0.00340), which means the prediction error is very small. So, overall, we can say that LSTM provides more accurate as well as more reliable results under different operating conditions.

The RNN model also gives good performance, with an R^2 value of **0.8947 (accuracy: 89.47%)**. Its RMSE and MAE values are also low, although they are slightly higher compared to LSTM. This shows that RNN is also capable of handling this type of problem quite well, but still, it is not as effective as LSTM when it comes to capturing long-term dependencies.

On the other hand, Random Forest and DNN show comparatively lower performance. Random Forest gives an R^2 value of **0.8312 (accuracy: 83.12%)**, while DNN gives **0.8266 (accuracy: 82.66%)**. These values are clearly lower than RNN and LSTM. The main reason behind this is that these models do not naturally capture time-based or sequential behavior. Since AGC signals depend heavily on past values, these models are not able to fully represent the system dynamics.

The better performance of LSTM can be explained by its ability to retain past information for a longer duration. In AGC systems, previous values such as frequency deviation (df1), RoCoF, and generator output have a strong influence on present control actions. Since LSTM can effectively learn and remember these past patterns, it produces more accurate predictions.

In addition to this, the use of lag features like df1_lag1, df1_lag2, RoCoF1_lag1, and GeneratorMW_lag5 further helps the model to understand the past behavior of the system. This again improves the overall prediction accuracy and model performance.

From all these observations, it becomes quite clear that models which can understand time-dependent behavior, like RNN and LSTM, are more suitable for AGC-related problems compared to basic machine learning models.

From a practical point of view, this is very important. In situations such as communication failure or cyberattacks, when the original AGC signal is not available, the system still needs a reliable backup input. In such cases, the LSTM model can generate a backup setpoint which is very close to the actual one, thereby helping in maintaining system stability and proper frequency control.

Finally, it can be concluded that the proposed machine learning-based decentralized approach works effectively. It provides a reliable backup mechanism, and among all the models considered, **LSTM is clearly the most suitable and best-performing model** for ensuring continuous and stable generator operation when centralized control signals are not available.

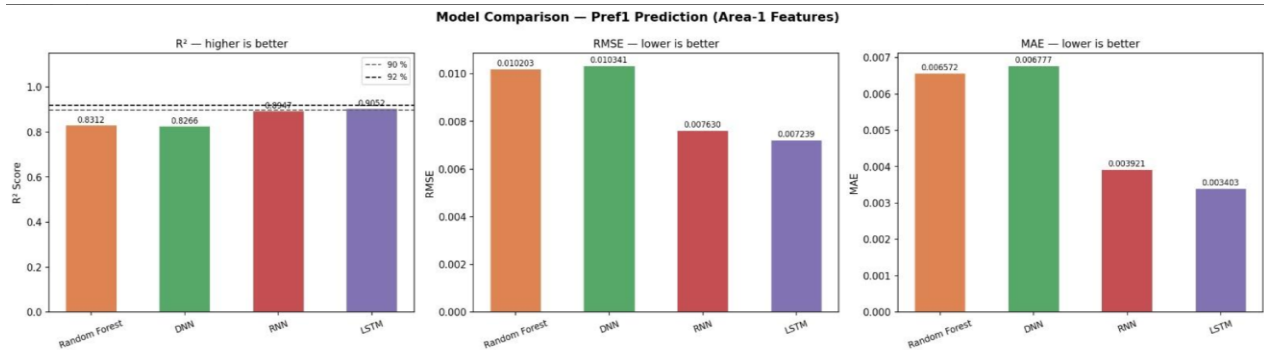


Fig.9 Model Comparison

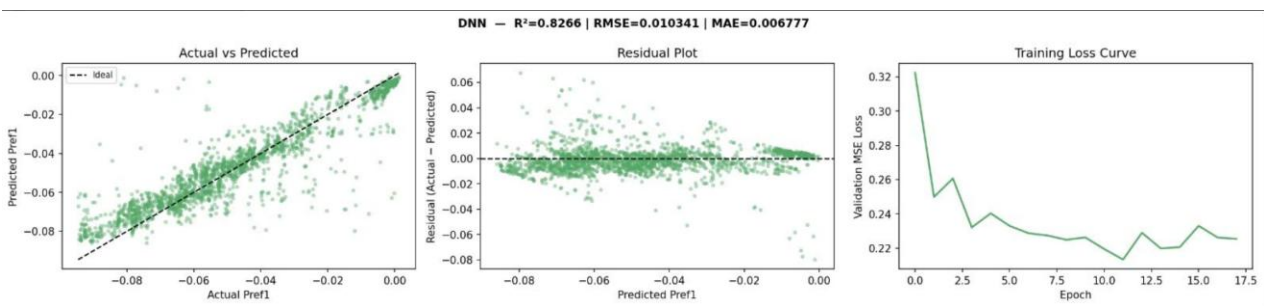


Fig.10 Deep Neural Network

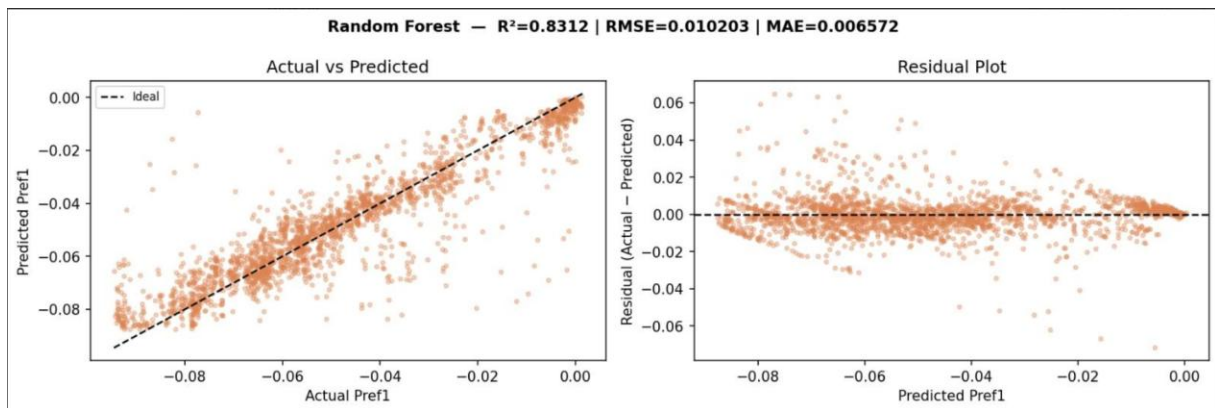


Fig.11 Random Forest

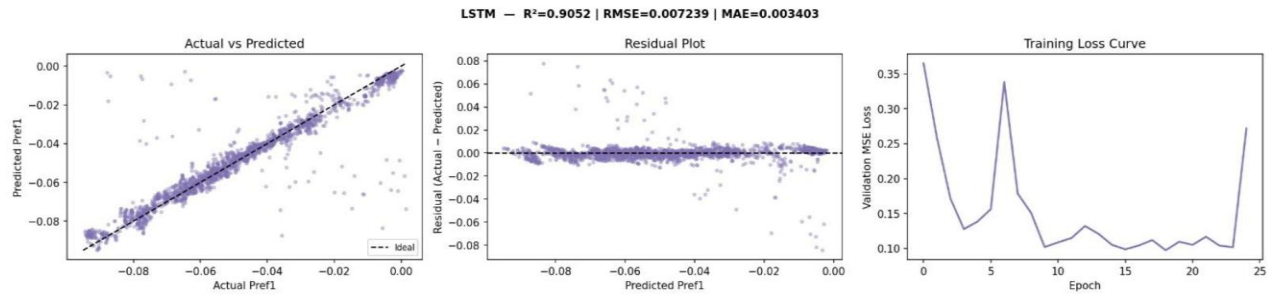


Fig.12 Long Short -Term Memory

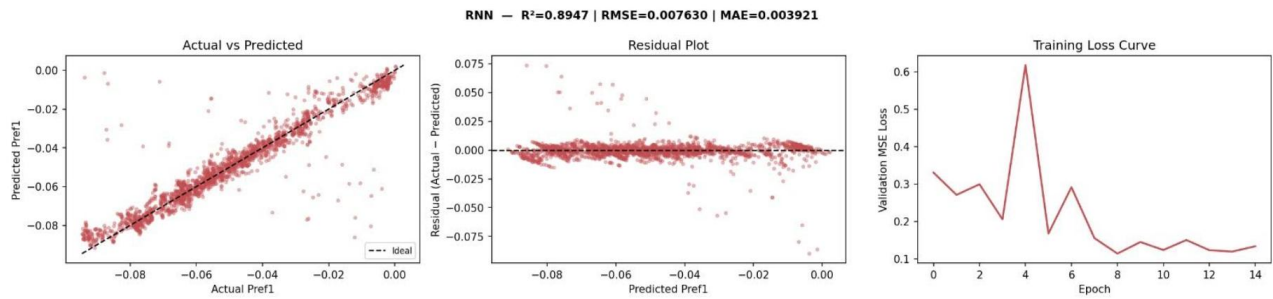


Fig.13 Recurrent Neural Network

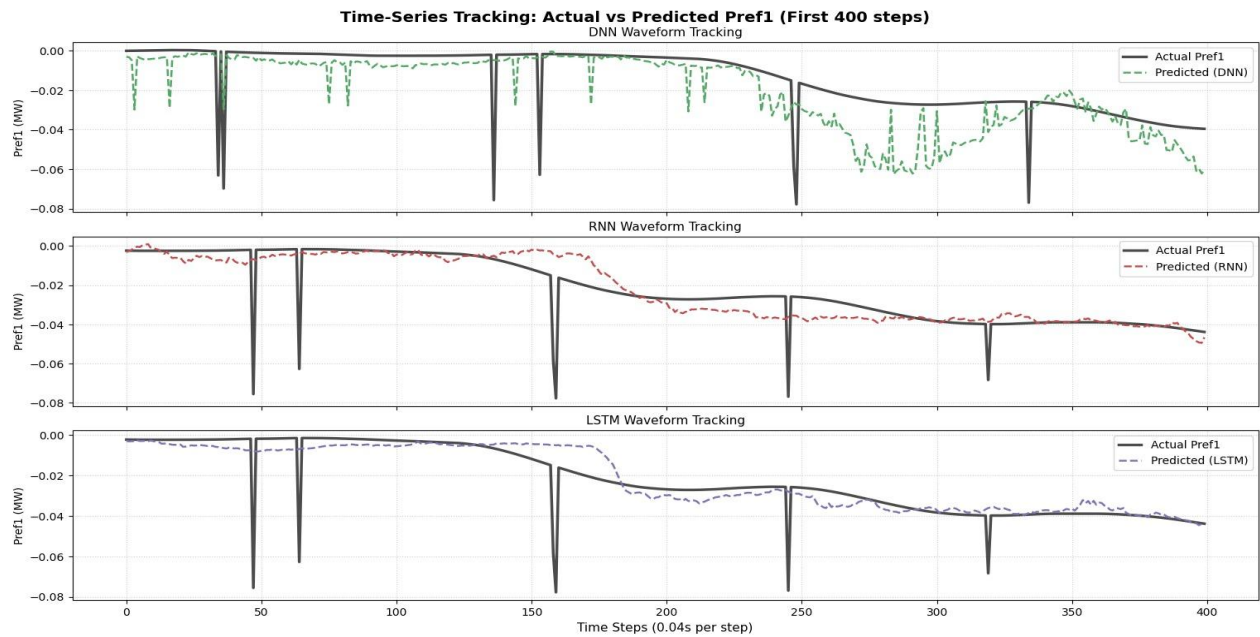


Fig.14 Time Series Tracking

Chapter - 5: APPLICATIONS

The ML-based decentralized backup setpoint framework and the underlying LSTM cyber-resilience techniques have broad applicability across modern cyber-physical power systems.

5.1 Post-Cyberattack AGC Recovery

The primary application is enabling uninterrupted generator participation in frequency regulation following a confirmed cyberattack on the AGC communication infrastructure. When an **FDI (False Data Injection)** or **TD (Time Delay)** attack is detected and the setpoint stream is suspended pending investigation and restoration, the backup setpoint model allows generators to maintain smooth control action rather than defaulting to a fixed setpoint or going offline. This directly reduces the risk of frequency instability and load shedding during the post-attack recovery window.

5.2 Communication Outage Resilience

The same backup setpoint mechanism provides operational continuity under non-malicious communication failures including packet loss, excessive propagation delay, control centre software faults, and maintenance-window unavailability. The decentralized nature of the local plant model ensures that AGC participation is maintained even when connectivity to the central control centre is disrupted, improving overall system reliability.

5.3 Smart Grid and Microgrid Applications

As power systems evolve toward smart grid architectures with increased distributed generation and microgrids, the volume and diversity of AGC communication signals increases substantially. The LSTM-based anomaly detection framework scales to multi-area systems with multiple tie-lines and can be adapted for microgrid frequency control applications where cyber security and communication robustness are equally critical concerns.

5.4 Renewable Energy Integration

The increasing penetration of variable renewable generation (wind and solar) introduces additional uncertainty and variability into AGC frequency regulation. The ML-based backup setpoint model, trained on historical AGC data that captures variable renewable conditions, can provide more adaptive setpoint estimates than fixed-parameter backup strategies during periods of high renewable variability, enhancing grid frequency stability in high-renewable scenarios.

5.5 Energy Management and SCADA Security

The detection and mitigation framework contributes to the broader goal of securing Energy Management System (EMS) and SCADA communication infrastructure against cyber-physical attacks. The framework's ability to detect and classify attack types, identify the attacked signal, and reconstruct corrected measurements in real time provides actionable information to system operators for cyber incident response and forensic analysis.

Chapter - 6: CONCLUSION

This project focused on developing a machine learning-based decentralized backup setpoint approach for generators operating under Automatic Generation Control (AGC). The main idea was simple but important—when centralized AGC signals are not available due to communication failure, delay, or cyberattacks, the system should still continue operating smoothly. To achieve this, different machine learning models such as Random Forest, DNN, RNN, and LSTM were tested to predict reliable backup setpoints based on system behavior.

From the results, it was clearly observed that models which understand time-dependent behavior perform better for this application. The LSTM model gave the best performance with an accuracy of **90.52%**, along with the lowest error values, making its predictions closest to actual system values. The RNN model also performed well with **89.47%** accuracy, but it was slightly less effective in capturing long-term dependencies. In comparison, Random Forest (**83.12%**) and DNN (**82.66%**) showed lower performance, mainly because they do not naturally capture the sequential nature of AGC data. This clearly shows that past system values play a very important role in predicting current control actions.

Overall, the proposed approach proves to be effective and practical. It provides a reliable backup mechanism that allows generators to continue operation even when centralized control signals are not available. This helps in maintaining frequency stability and avoiding sudden disturbances in the system. Among all the models, **LSTM stands out as the most suitable and reliable choice**, making it a strong solution for improving the resilience and reliability of modern power systems.

Chapter - 7: FUTURE WORK

Several directions for extending this work are proposed:

- **Markov Decision Process (MDP) Attack Modelling:** The attacker's decision process will be formulated as a finite-horizon MDP for limited knowledge scenarios and an infinite-horizon MDP for scenarios where the attacker can anticipate future system states, enabling the operator to calculate attack likelihoods and assess specific component vulnerabilities.
- **Moving Target Defence (MTD):** MTD techniques will be investigated as a complementary mitigation approach that alters the attack surface dynamically, making it more difficult for adversaries to maintain a stable FDI attack strategy.
- **Multi-Area and Multi-Attack Coordination:** The risk analysis framework will be extended to cover coordinated attacks targeting multiple AGC signals simultaneously and to multi-area systems with more complex tie-line topologies.
- **Differentiation of Attacks from Physical Contingencies:** The attack-resilient control framework will be enhanced to distinguish abnormal measurements caused by cyber-attacks from legitimate measurement deviations caused by power system contingencies such as generator trips and line faults.
- **Hardware-in-the-Loop Validation:** Real-time hardware-in-the-loop testing of the detection and backup setpoint framework on a power system simulator will be pursued to validate performance under realistic communication channel conditions.

Chapter - 8: REFERENCES

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